

ORLANDO MELBOURNE INTL, FL, USA

WMO: 722040

Lat: 28.101N

Lon: 80.644W

Elev: 8

StdP: 101.23

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 12838

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.7	(c) 6.2	(d) -3.9	(e) 2.7	(f) 8.1	(g) -1.1	(h) 3.5	(i) 9.6	(j) 11.1	(k) 17.6	(l) 9.7	(m) 16.9	(n) 3.5	(o) 310	(p) 0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.6	33.2	25.4	32.5	25.4	32.0	25.5	26.9	31.0	26.6	30.7	26.1	30.2	4.7	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.9	21.3	29.0	25.2	20.4	28.7	25.0	20.1	28.5	83.8	30.8	82.2	30.5	80.9	30.4	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.3	(b) 8.4	(c) 7.8	(d) DB	(e) 0.4	(f) 35.5	(g) 2.0	(h) 1.1	(i) -1.0	(j) 36.3	(k) -2.2	(l) 36.9	(m) -3.4	(n) 37.5	(o) -4.8	(p) 38.3	(q) 30.7
			(d) WB	(e) -1.4	(f) 28.1	(g) 2.0	(h) 1.0	(i) -2.9	(j) 28.8	(k) -4.0	(l) 29.4	(m) -5.2	(n) 30.0	(o) -6.6	(p) 30.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.1	16.5	18.1	20.0	22.4	25.1	27.1	27.8	27.9	27.3	24.8	20.8	18.5		
	DBStd	5.01	4.57	4.38	3.76	2.83	1.89	1.38	1.18	1.15	1.29	2.80	3.55	4.48		
	HDD10.0	12	7	2	0	0	0	0	0	0	0	0	0	3		
	HDD18.3	251	88	53	29	5	0	0	0	0	0	3	18	56		
	CDD10.0	4778	210	230	309	373	468	513	552	556	518	458	326	267		
	CDD18.3	1975	33	47	79	128	210	263	293	297	268	202	93	61		
	CDH23.3	17262	134	223	408	843	1764	2543	3081	3132	2682	1710	497	246		
(14) Wind	WSAvg	3.9	3.9	3.9	4.3	4.4	4.3	3.5	3.2	3.3	3.6	4.2	4.0	3.7		
	PrecAvg	1318	58	52	77	59	85	171	158	188	180	135	58	61		
	PrecMax	1770	154	168	297	147	257	363	297	610	361	290	206	229		
	PrecMin	962	0	1	0	16	17	3	55	37	88	19	6	10		
	PrecStd	244	35	36	73	33	58	81	65	110	73	79	44	55		
	DB	27.9	29.4	31.1	32.0	33.4	34.7	34.1	34.0	33.3	31.7	29.7	28.4			
	MCWB	21.3	21.1	21.7	22.1	23.4	25.3	25.8	25.7	25.2	25.1	23.6	22.6			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	26.3	27.5	28.7	29.8	31.7	32.9	32.9	32.8	32.1	30.5	28.0	27.2			
	2% MCWB	20.7	21.1	21.2	22.5	23.2	25.2	25.7	26.0	25.5	24.6	22.9	22.1			
	5% DB	24.8	26.1	27.2	28.3	30.2	32.0	32.3	32.2	31.2	29.5	27.2	26.1			
	5% MCWB	20.0	20.6	20.8	21.8	23.3	25.3	25.7	26.0	25.3	24.2	22.3	21.5			
	10% DB	23.2	24.4	25.9	27.4	29.0	31.0	31.4	31.3	30.6	28.7	26.2	24.7			
	10% MCWB	19.4	19.9	20.0	21.2	23.0	25.0	25.6	25.8	25.1	23.7	21.3	20.9			
	WB	22.9	23.0	23.8	24.5	25.3	26.9	27.3	27.4	27.1	26.5	25.0	24.0			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% MCDB	26.1	26.2	27.2	28.6	29.9	31.8	31.6	31.3	30.6	29.4	27.7	26.8			
	2% WB	21.9	22.2	22.8	23.7	24.7	26.2	26.7	26.9	26.6	25.7	24.1	23.1			
	2% MCDB	24.4	25.6	26.1	27.5	29.1	31.0	31.2	30.9	30.1	28.7	26.9	25.6			
	5% WB	21.1	21.5	21.9	23.0	24.2	25.7	26.3	26.6	25.9	25.2	23.3	22.5			
	5% MCDB	23.6	24.8	25.5	26.6	28.4	30.2	30.9	30.6	29.5	28.3	26.1	24.9			
	10% WB	20.2	20.7	21.1	22.4	23.8	25.4	25.8	26.2	25.6	24.7	22.5	21.7			
	10% MCDB	22.8	23.9	24.6	26.0	27.8	29.6	30.2	30.1	29.1	27.7	25.2	24.0			
(35) Mean Daily Temperature Range	MDBR	10.5	10.1	9.9	9.2	8.4	7.8	7.9	7.6	7.0	7.3	8.8	9.3			
	5% DB	10.1	10.0	10.3	9.9	10.1	9.4	8.7	8.2	7.6	7.2	8.1	8.6			
	5% MCWBR	5.6	4.9	4.8	4.4	4.1	3.8	3.5	3.3	3.1	3.1	4.2	4.7			
	5% WB	8.9	8.9	9.1	8.6	8.6	8.2	8.0	7.6	7.1	6.4	7.2	7.6			
	5% MCWBR	5.6	4.9	4.7	4.3	3.9	3.7	3.6	3.5	3.2	3.0	4.1	4.7			
(40) Clear-Sky Solar Irradiance	taub	0.346	0.359	0.373	0.396	0.427	0.454	0.472	0.461	0.426	0.401	0.364	0.353			
	taud	2.532	2.509	2.458	2.373	2.328	2.328	2.276	2.325	2.415	2.452	2.504	2.535			
	Ebn at Noon	899	914	918	901	868	839	824	832	855	858	869	872			
	Edh at Noon	90	99	110	123	129	129	135	128	114	104	92	86			
(44) All-Sky Solar Radiation	RadAvg	3.40	4.13	5.15	6.08	6.25	5.87	5.86	5.59	4.92	4.38	3.58	3.09			
	RadStd	0.24	0.24	0.33	0.24	0.45	0.53	0.33	0.34	0.29	0.28	0.18	0.21			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+182	+158
(47) Regional (31 neighbors)		+0.43	N/A	N/A	N/A	N/A	N/A	N/A	-41	+157	+114

Nomenclature: See separate page